

Part V

Calibration and the Recalibration Application

Calibrating to Caplets

16.1 From market quotes to model parameters: the pipeline

Everything in this book so far has been building toward one procedure: given today's live cap, floor and swaption prices, find the parameters — $\sigma_i(t)$ (or the Rebonato a, b, c, d that generate it), ρ_{ij} (or the correlation-form parameters that generate it), and any displacement c_i — that make the BGM model reproduce those prices. This is a two-stage process, addressed in this chapter and the next: caplets first, then swaptions, and the reason for that order is that caplet vols enter the swaption-matching problem of Chapter 17 as an input, not an output.

Key result: the calibration pipeline

1. **Strip** flat market cap/floor quotes into per-caplet (“spot”) volatilities (this chapter).
2. **Fit** a parametric instantaneous volatility form (Chapter 14's $\sigma_i(t)$) to the stripped caplet vols.
3. **Calibrate** correlation and any remaining free volatility degrees of freedom jointly to the swaption market (Chapter 17).
4. **Validate**: reprice every input instrument with the fitted model and check the residuals are within tolerance (Chapter 18 and 20).

16.2 Stripping caplet vols from flat cap quotes

Chapter 6 flagged the mismatch: the market quotes one **flat** volatility per whole cap (e.g. the 5-year cap), applied uniformly to every caplet inside it, while BGM needs a genuinely different volatility for each individual caplet (the **spot** or **caplet** vol). Since a cap's price is the sum of its caplet prices, and each caplet price is individually monotonic in its own volatility, the caplet vols can be recovered by working outward from the shortest cap.

Key result: caplet stripping

Let $C(K, T_m)$ denote the market price of the cap with strike K and m caplets, quoted at flat vol σ_m^{flat} . Since

$$C(K, T_m) = \sum_{i=1}^m \text{Caplet}_i(\sigma_m^{\text{flat}}) = \sum_{i=1}^{m-1} \text{Caplet}_i(\sigma_i^{\text{spot}}) + \text{Caplet}_m(\sigma_m^{\text{spot}})$$

if $\sigma_1^{\text{spot}}, \dots, \sigma_{m-1}^{\text{spot}}$ are already known from stripping shorter caps, the only unknown is σ_m^{spot} , recovered by solving

$$\text{Caplet}_m(\sigma_m^{\text{spot}}) = C(K, T_m) - \sum_{i=1}^{m-1} \text{Caplet}_i(\sigma_i^{\text{spot}})$$

a single-variable root-find (Newton–Raphson converges in a handful of iterations, since a caplet's price is smooth and monotonic in its own volatility) at each successive maturity,

in order from shortest to longest.

Worked example 16.1 — stripping a two-caplet toy cap ladder.

Simplify to a two-period world: a 1-caplet cap (period $[0.5, 1.0]$) trades flat at $\sigma_1^{\text{flat}} = 88$ bp normal, priced at \$4,120 for a \$10mm notional. A 2-caplet cap (periods $[0.5, 1.0]$ and $[1.0, 1.5]$) trades flat at $\sigma_2^{\text{flat}} = 91$ bp normal, priced at \$9,430 total. Find the spot vol for the second caplet.

Step 1 — the first caplet's spot vol equals its flat vol. With only one caplet in the shortest cap, there is nothing to strip: $\sigma_1^{\text{spot}} = 88$ bp, and its price under this vol is exactly the quoted \$4,120 by definition.

Step 2 — isolate the second caplet's target price.

$$\text{Caplet}_2 \text{ target} = \$9,430 - \$4,120 = \$5,310$$

Step 3 — solve for σ_2^{spot} . Given the second caplet's forward, strike, fixing time and discount factor (held fixed, known from the curve), a Newton–Raphson search finds the normal volatility that reprices this caplet at exactly \$5,310; suppose this converges to $\sigma_2^{\text{spot}} = 94$ bp.

Sanity check. 94 bp > 91 bp (the flat vol) > 88 bp (the first caplet's spot vol) is a sensible pattern here: the flat vol is a kind of weighted average of the two spot vols, so if it rose from the 1-caplet to the 2-caplet cap, the newly added caplet's own vol should generally sit above the blended average, not below it.

Interpretation. 94 bp is the volatility Chapter 14's $\sigma_i(t)$ parametric form must be fit to reproduce, specifically for the second caplet's own time-to-expiry — not the 91 bp headline quote, which is a blend across two different caplets with two different underlying volatilities and should never itself be fed directly into a single-caplet pricing formula.

PITFALL

A vega-weighted, rather than price-weighted, view of the stripping problem matters once real market data (with bid/ask spreads and quoting noise) replaces the toy example above: a caplet deep out-of-the-money contributes very little vega to its cap's total price, so a strip that matches cap *prices* exactly can still leave the implied spot vol for a low-vega caplet wildly under-determined — small pricing noise in the input cap quotes translates into large, unstable swings in that specific caplet's stripped vol. Production stripping algorithms typically vega-weight the fit, or apply light regularization (e.g. preferring a smooth spot-vol curve over successive maturities) specifically to prevent this instability, rather than treating the strip as a sequence of independent, unregularized root-finds.

16.3 Fitting the parametric form

Once a full ladder of spot (caplet) vols is available, Chapter 14's Rebonato parameters a, b, c, d are fit by a straightforward nonlinear least-squares regression: minimise

$$\sum_i w_i \left(\sigma_i^{\text{spot}} - \sigma_i^{\text{Rebonato}}(a, b, c, d) \right)^2$$

over a, b, c, d , with weights w_i typically set by vega or by liquidity (more liquid maturities weighted more heavily, since their stripped vol is more reliable). This is a small, well-behaved,

four-parameter nonlinear optimization — fast, and rarely a source of numerical difficulty on its own, in contrast to the correlation calibration of the next chapter.

ON THE DESK

On a live desk, the caplet stripping and parametric fit described in this chapter typically runs first, every morning, before anything touches the swaption market at all, because the caplet vol curve is also used directly — independent of BGM entirely — to mark and risk-manage the vanilla cap and floor book. A recalibration engine that structures its pipeline this way gets a valuable side benefit: the caplet stripping step alone provides an early, cheap sanity check on the day's market data (a caplet vol that suddenly jumps by 50 bp between two adjacent maturities, with nothing macro to explain it, is far more likely to be a bad quote than a genuine market move) before the more expensive joint swaption calibration of the next two chapters is even attempted.

Chapter benchmark

Before moving on, you should be able to, without notes:

1. Explain the difference between a flat cap vol and a spot (caplet) vol, and why BGM needs the latter.
2. Describe the caplet-stripping recursion: solve shortest to longest, each step isolating exactly one new unknown.
3. Explain why vega-weighting (or regularization) matters for stripping low-vega, deep out-of-the-money caplets, and what goes wrong without it.

